

# Image and Voice Pipelines

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of speech pipelines and multimodal evidence.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

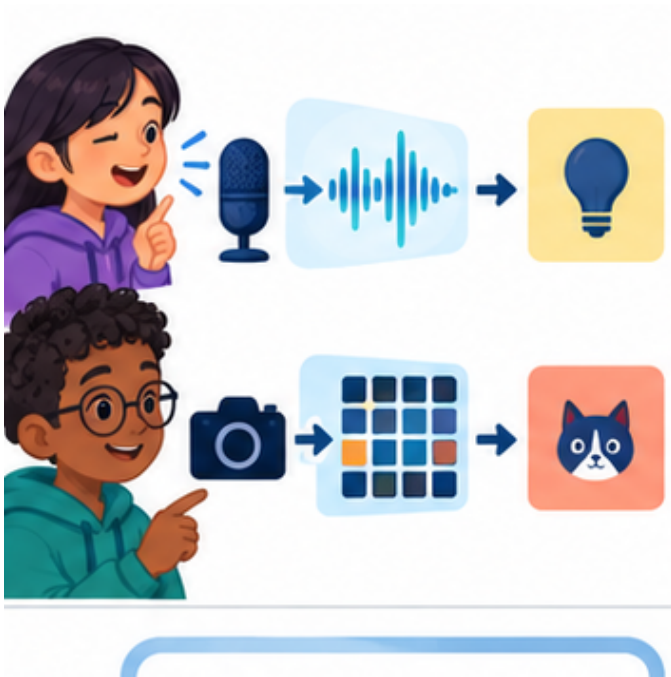
Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 26: Image and Voice Pipelines

Perception systems transform signals through multiple fallible stages.



### THINKER MISSION

Trace speech-to-command and image-to-label pipelines.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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### Readiness for the connected explanation

Recall Image generation and provenance. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

### This week's end goal

By the end, I can trace where speech information changes and resolve a declared-authority conflict.

## 2. Notice the evidence

### PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use image and voice pipelines to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

**Write or draw your reasoning and evidence.**

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My prediction and the clue supporting it:

**Write or draw your reasoning and evidence.**

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## A closer explanation

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

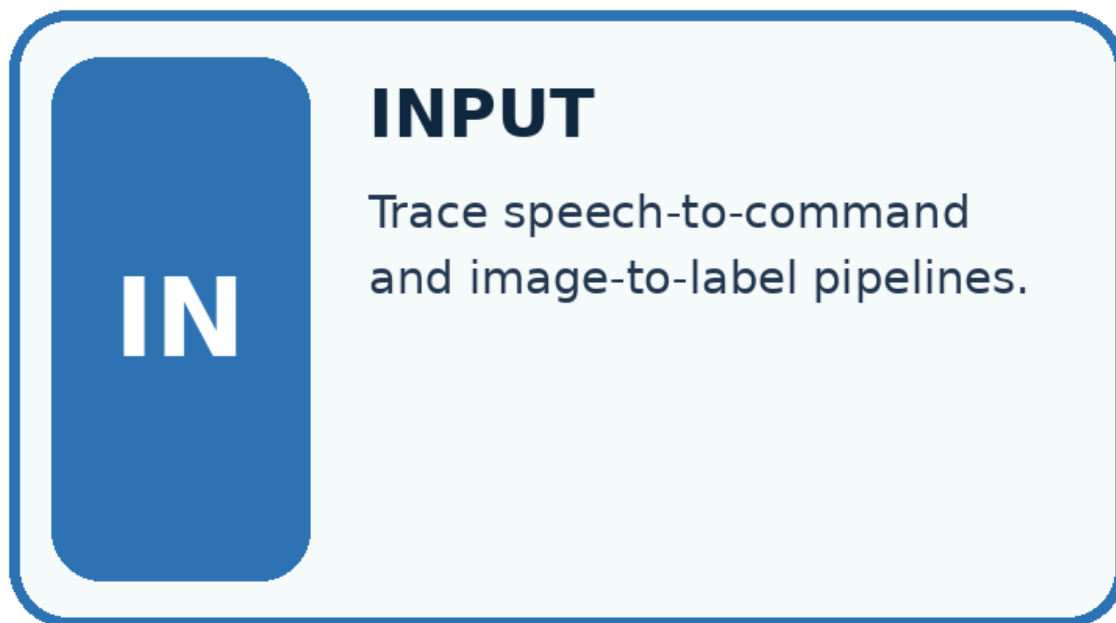
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

## 3. See how it works

### VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## PROCESS

Perception systems transform signals through multiple fallible stages.





## OUTPUT

Trace speech-to-command and image-to-label pipelines.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

## Speech pipelines and multimodal evidence — evidence trace

### Input and scope · Check or calculation · Conclusion and limit

**Input and scope:** Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

**Check or calculation:** Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

**Conclusion and limit:** There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

# 4. Compare the cases

## CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



**NORMAL**  
Expected everyday case



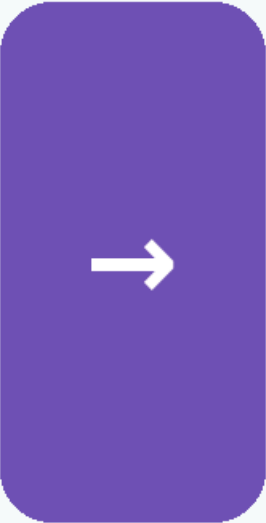
**BOUNDARY**  
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## MISSING

Needed information absent



## TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: Remove access to the audio, while keeping the authoritative inventory.

Prediction: Shelf 14 remains supported for the lookup, but transcription accuracy cannot be replay-checked.

Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

**SIGNAL**  
measured sound or light  
values

2

## **PIPELINE**

ordered stages transforming information

3

## **RECOGNITION**

convert signal into a symbol or label

4

## STAGE ERROR

failure introduced at one step

**Word · My own example · Not this / boundary**

**Word:** signal

**My own example:** A measurable quantity carrying information, such as changing sound pressure or light intensity.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** pipeline

**My own example:** A connected sequence of stages in which one stage's output becomes another's input.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** recognition

**My own example:** Interpreting an input as a known category or pattern rather than merely recording its signal.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** stage error

**My own example:** A failure introduced at a particular step of a pipeline.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Modality

**Meaning:** A form of information such as audio, image or text

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**Word:** Transcription

**Meaning:** A text representation of speech

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**Word:** Error propagation

**Meaning:** An earlier error affecting later processing

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## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 26 • WORKED EVIDENCE

- 1 Voice: sound → sampled signal → recognized words → intent → action.
- 2 Image: pixels → features → score → label.
- 3 Noisy sound can produce wrong words even if the intent stage works.
- 4 Diagnose the first wrong stage before changing later stages.

1. Voice: sound → sampled signal → recognized words → intent → action.

2. Image: pixels → features → score → label.

3. Noisy sound can produce wrong words even if the intent stage works.

4. Diagnose the first wrong stage before changing later stages.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Image and Voice Pipelines

A fictional librarian dictates a shelf note that passes through transcription and catalogue lookup.

### The facts we will use

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

## Image and Voice Pipelines

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Audio input

Known phrase names fourteen, not forty.



#### 2. Transcription

Text output changes the shelf number to forty.



#### 3. Cross-check

The authoritative inventory gives shelf 14.



#### 4. Review and correct

Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

### CHANGE ONE THING

Remove access to the audio, while keeping the authoritative inventory.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Audio input

**What happens and why:** Known phrase names fourteen, not forty.

**Step:** 2. Transcription

**What happens and why:** Text output changes the shelf number to forty.

### Step: 3. Cross-check

**What happens and why:** The authoritative inventory gives shelf 14.

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### Step: 4. Review and correct

**What happens and why:** Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

A correct later lookup cannot repair an earlier wrong input by magic. Locate the first difference between known audio and transcript, then compare the stated source authority. Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

## What this example does not establish

Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

Read the labelled picture: Which stage first fails, and why should the faulty transcript not be silently overwritten?

Change one thing: Remove access to the audio, while keeping the authoritative inventory. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

### ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: The final model is the only possible source of a wrong output. Strong move: locate evidence, change one cause, and retest.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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## 8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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## A closer explanation

A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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## Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Speech pipelines and multimodal evidence”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Speech pipelines and multimodal evidence. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

**Explain the picture**

Which stage first fails, and why should the faulty transcript not be silently overwritten?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

**9. Run the investigation**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



**PREDICT → TEST → RECORD → EXPLAIN**

Trace four failures backward through stage cards. Change one stage input, predict downstream effects, and propose a stage-specific test.

Trace four failures backward through stage cards. Change one stage input, predict downstream effects, and propose a stage-specific test.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



**MASTERY LADDER**  
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

**Write or draw your reasoning and evidence.**

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## 11. Connect your project

### CUMULATIVE PROJECT

#### Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



**TRANSPARENT BOOK RECOMMENDER**  
If voice or image input is added, show each transformation and a safe fallback.

**THIS WEEK'S CHECKPOINT** If voice or image input is added, show each transformation and a safe fallback.

**SYSTEM MAP** Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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**RESPONSIBLE DESIGN** Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

## Evidence for your project

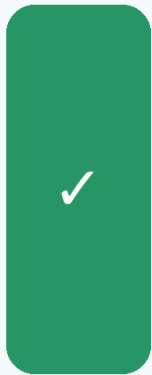
For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

# 12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



## INDEPENDENT PROOF

Trace speech-to-command and image-to-label pipelines.

**CHALLENGE** Trace speech-to-command and image-to-label pipelines.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

A voice book helper transcribes speech, extracts preference, then ranks books.

#### The question

Where can a wrong suggestion begin?

#### Worked explanation

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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**Trace the supplied case visually**

**Locate the first error in a voice pipeline**

**Evidence or stage · Value or result**

**Evidence or stage:** Speech

**Value or result:** Reader says poetry

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**Evidence or stage:** Transcript

**Value or result:** May mishear the preference

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**Evidence or stage:** Extract

**Value or result:** Uses the transcript's genre

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**Evidence or stage:** Rank

**Value or result:** Wrong input can produce a wrong suggestion

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A voice book helper transcribes speech, extracts preference, then ranks books.

Where can a wrong suggestion begin?

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

## 14. Try a new case independently

**Independent case: Apply a new case independently**

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the transcription limit.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

**Week 26 Independent application**

Close the worked explanation. Apply the idea to this different question without copying.

A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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**Create and test**

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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**Connect to your project**

If voice or image input is added, show each transformation and a safe fallback.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

**Transfer practice from the original programme**

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional librarian dictates a shelf note that passes through transcription and catalogue lookup.

### By the end, I can · How I will show it

**By the end, I can:** Explain the mechanism and terminology of speech pipelines and multimodal evidence

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Apply the stated procedure to a new case with a traceable result

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Identify an assumption or limitation and design a revealing follow-up

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of speech pipelines and multimodal evidence

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.